DESIGN

SOFTWARE ENGINEERING II

TESTIGOS DE MELUK

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System domain

System context

Industrias Escolares is a school furniture manufacturing company (chairs, tables, and desks) with a workforce of 10 to 20 employees. In its current setup, the owner provides quotes directly and generates the work order once the request is ready, which is subsequently invoiced. This entire process is recorded in the desktop version of Siigo Pyme, installed on a single, non-shared computer. This dynamic defines the central restriction of the new design: the proposed system must coexist with Siigo without replacing it or writing to it automatically, as there is no public API and tax accounting will remain the exclusive responsibility of the accounting software.

The actual operational flow that the new system must support remains intact: the owner provides quotes, generates the work order, and the secretary logs it into Siigo. The integration of a virtual assistant will solely serve to feed the initial phase with preliminary quotes, without replacing anyone in the current workflow. A key factor at this stage is that there are no official prices, nor will they be displayed in the product catalog. Instead, preliminary price ranges will be estimated dynamically by the virtual assistant based on historical quotes; during the initial phases or when no prior quotes exist, the bot will extract pricing data from past WhatsApp conversations to determine these ranges. The final exact price depends primarily on customer tenure and order volume, variables that are adjusted based on workshop workload, raw material costs, material availability, and transportation.

Finally, the physical environment heavily dictates the security design. The computer hosting Siigo is located facing the door, leaving information exposed. Data backup is performed manually on a single USB drive that only management connects at closing time, and there is a Mac computer where company and personal files are mixed across two USB drives. Add to this that the network runs exclusively on Wi-Fi with no guaranteed hardwiring, and the total hardware inventory is limited to two desktop computers and one laptop. This highlights that the security problem transcends the digital realm and includes physical vulnerabilities; consequently, the new system's security module must combine strict account and session control with a verifiable backup discipline.

Characterization of actors

Actor	Type	Characterization
Owner / Management	Internal user, Administrator role	Provides quotes directly without intermediaries, generates work orders (W.O.), approves or rejects quotes generated by the assistant, and determines final prices with proper justification. She is the only person authorized to connect the backup USB drive and manage the DIAN (tax authority) password, although the assistant types it in.

		Additionally, she makes all decisions regarding production schedule saturation.
Secretary / Accounting Assistant	Internal user, Operator role	Responsible for logging the information generated by management (such as work orders) into Siigo, as well as importing data and viewing customer profiles. Operates the shared computer, which is subject to an automatic lock due to inactivity.
Accounting	Internal user, Operator role	Primarily responsible for reviewing financial reports and accessing customer profiles.
Potential Customer	External user, no account	Can request quotes 24/7 through the virtual assistant using a guided form (requiring name, email, and phone number). Receives a preliminary quote in PDF format or via email. Under no circumstances do they have access to internal pricing or system logs.
Super Administrator	System Administrator	Responsible for creating and deactivating accounts, with mandatory role assignment. Also manages the backup inventory and performs restoration tests in an isolated environment (sandbox).
Siigo Pyme	External system, non-human	The authoritative accounting software, installed on a single computer with no public API. It receives data via manual uploads from files exported by the new system. It operates as a one-way workflow, with no automatic data entry or real-time synchronization.
Backup Service	Subsystem, non-human	Executes differential backups daily at closing time. Maintains a retention scheme that includes 7 daily, 4 weekly, and 12 monthly backups. Furthermore, it performs data integrity checks, issues reminders, and manages a second physical hard drive for rotation.

Business rules

Business Rules for Module 1 (Security and Backups)

- **RN-M1-01:** All user accounts are individual and require a mandatory role assignment (Super Administrator, Administrator, or Operator). Accounts cannot be created

without a defined role, and system permissions depend exclusively on this role, never on the specific individual.

- **RN-M1-02:** No session can operate without two-factor authentication. Once a valid password is provided, the system requires a six-digit temporary code before issuing access tokens. Three consecutive failed attempts will temporarily lock the account.
- **RN-M1-03:** Only one active session is allowed per account. A new login automatically invalidates the previous one. Additionally, after fifteen minutes of inactivity, the session is automatically closed on the shared computer.
- **RN-M1-04:** The system performs a daily data backup at closing time, maintaining a rotation scheme that keeps seven daily, four weekly, and twelve monthly copies. This process includes a data integrity check and triggers an alert if the USB drive is not connected. Any data restoration must first be tested in an isolated sandbox environment before being applied to production.
- **RN-M1-05:** Every critical action within the system, such as final quote approvals, role modifications, logins, or backup executions, is recorded in an unalterable audit log. This record details who performed the action, when it occurred, and the exact reason behind it.

Business Rules for Module 2 (Commercial and Assistant)

- **RN-M2-01:** Products in the catalog will not display any official, fixed, or reference prices. The catalog will function strictly as a repository of products and technical specifications, ensuring that pricing remains entirely dynamic and handled exclusively through the quoting process.
- **RN-M2-02:** Only management is authorized to determine and adjust the final exact price on individual quotes. This adjustment must be recorded in the audit log, detailing the preliminary price range estimated by the bot, the final approved price, the date, the responsible user, and the reason. Any attempt to approve or adjust quote prices by another role will be denied and logged in the audit trail.
- **RN-M2-03:** The virtual assistant is restricted to generating preliminary quotes by estimating price ranges dynamically based on previous system quotes; if no prior quotes exist, it will extract pricing data from historical WhatsApp conversations to formulate these ranges. This process is carried out through a guided form that prevents free-text entry, and the document is sent via the WhatsApp Business API with a disclaimer stating it is preliminary and subject to approval.
- **RN-M2-04:** No quote can advance in the workflow without human approval. The owner is responsible for approving, amending, or rejecting requests, always stating the reason. If a quote is rejected, it reverts to its preliminary status and never enters the export queue for Siigo.
- **RN-M2-05:** The estimated delivery date corresponds to the dispatch date, which is calculated using the historical median based on product and volume, while adding time buffers for potential absenteeism or material shortages. If there is insufficient historical data to perform the calculation, the system clearly reports this and refrains from fabricating a date.
- **RN-M2-06:** Every work order has a unique number and follows a sequential cycle: it starts as pending, is then logged into Siigo by the secretary, and is finally marked as reconciled. The document includes mandatory fields for polypropylene, paint, wood

details such as bevels, lacquered edges, description, and Formica, as well as unit values, subtotals, totals including VAT, and any applicable advance payments.

- **RN-M2-07:** Data is sent to Siigo exclusively through structured batch export files, generated manually by the secretary from approved work orders. Automatic data entry or real-time synchronization with the accounting software is strictly prohibited.
- **RN-M2-08:** The comprehensive 360-degree customer profile, grouped by Tax ID, consolidates contact information, commercial history, and private notes, which are only visible to authorized roles. Furthermore, it triggers a risk concentration alert when a customer exceeds an established purchasing threshold. VAT is always included in the displayed price.

Assumptions Dictating the Design

1. The automatic session timeout after fifteen minutes of inactivity is accepted to protect information on the shared computer.
2. Computers are turned off at night. Without an overnight operational window, batch processes must strictly run at the end of the workday.
3. The implementation of the WhatsApp channel alongside the guided form must be explicitly confirmed with management.
4. The brand and model of the fingerprint scanner, as well as the exact format of the physical work order, remain to be validated on-site.

CRC Cards

Module 1 (Security and Backups)

Role	
Responsibilities	**Collaborators**
• Define one of the three system roles (Super Administrator, Administrator, Operator). • Hold the set of permissions tied to the role, never to the person. • Answer whether the role grants a given permission.	• User

User	
Responsibilities	**Collaborators**
• Represent an individual employee account: full name, unique email and account status. • Be linked to exactly one mandatory role; change it when authorized. • Activate or deactivate the account and update the profile.	• Role • TOTPEnrollment • Session • AuditEntry

• Check permissions through its role and record its last access.	

TOTPEnrollment	
Responsibilities	**Collaborators**
• Store the secret that links the user with the authenticator app. • Know its expiration time (five minutes) and report whether it has expired. • Be confirmed or invalidated; a user has at most one.	• User

Session	
Responsibilities	**Collaborators**
• Represent a login session with its status and last activity time. • Update its last activity and close itself after 15 minutes of inactivity. • Be terminated when a new login replaces it (one active session per account). • Report whether it is still active.	• User

AuditEntry	
Responsibilities	**Collaborators**
• Record, in an unalterable way, who did a critical action, what it was, on which entity and when. • Be append-only: it can never be edited or deleted.	• User

BackupFile	
Responsibilities	**Collaborators**
• Represent one backup copy: type (daily, weekly or monthly), SHA-256 hash and integrity status. • Compare its stored hash with a computed one to verify integrity. • Mark the result of the integrity check.	• RestoreProcess

RestoreProcess	
Responsibilities	**Collaborators**
• Track each restoration attempt made from a backup. • Register the result of the test in the isolated sandbox before touching production. • Allow confirming and completing the restoration only if the verification passed.	• BackupFile • User

Module 2 (Commercial and Virtual Assistant)

Product	
Responsibilities	**Collaborators**
• Represent a catalog item: unique code, name and status. • Hold no official price, since there is no up-to-date price list: reference prices and the suggested range come from historical sales data. • Be activated or deactivated and tell whether it can be selected in a quotation or work order.	• QuotationItem • WorkOrderItem

Customer *«abstract»*	
Responsibilities	**Collaborators**
• Hold the data shared by every customer: tax ID, contact information and status. • Group the 360-degree profile: contact, commercial history and private notes. • Let subclasses define how the customer name is displayed and its type. • Be activated or deactivated.	• NaturalPerson • LegalEntity • CommercialNote • Quotation • WorkOrder

NaturalPerson *extends Customer*	
Responsibilities	**Collaborators**

• Represent an individual customer identified by their ID document number. • Provide the person's full name as the display name.	• Customer

LegalEntity *extends Customer*	
Responsibilities	**Collaborators**
• Represent a company customer identified by its NIT. • Provide the business name as the display name.	• Customer

CommercialNote	
Responsibilities	**Collaborators**
• Store a private follow-up note about a customer with its content and author. • Be edited only by authorized roles. • Feed the customer's 360-degree profile.	• Customer • User

Quotation	
Responsibilities	**Collaborators**
• Represent a quotation: customer (optional for potential customers), items and an estimated total expressed as a price range, never a fixed amount. • Recalculate the estimated price range when items are added or removed. • Be approved, modified or rejected by the owner, always stating the reason when rejected. • Never advance in the workflow without human approval.	• QuotationItem • Customer • ProcessingQueueEntry • User

QuotationItem	
Responsibilities	**Collaborators**
• Represent one quotation line: product, quantity and the unit price range at the time of quoting. • Calculate its estimated subtotal range and allow changing quantity and price range.	• Quotation • Product

WorkOrder	
Responsibilities	**Collaborators**
• Represent a work order with a unique sequential number, customer, items and agreed delivery date. • Follow its cycle: pending, logged in Siigo, reconciled. • Add, update and remove items, reschedule the delivery date and be cancelled with a reason.	• WorkOrderItem • Customer • Product • User

WorkOrderItem	
Responsibilities	**Collaborators**
• Represent one work order line: product and quantity to manufacture. • Allow changing the quantity.	• WorkOrder • Product

ProcessingQueueEntry	
Responsibilities	**Collaborators**
• Represent an approved quotation waiting to be exported to Siigo. • Keep its export status and the time it was enqueued. • Be marked as exported by a batch; rejected quotations never enter the queue.	• Quotation • ExportBatch

ExportBatch	
Responsibilities	**Collaborators**
• Represent the batch file generated manually by the secretary. • Group one or more queue entries into a single structured file to upload into Siigo. • Know who created it and how many orders it contains.	• ProcessingQueueEntry • User

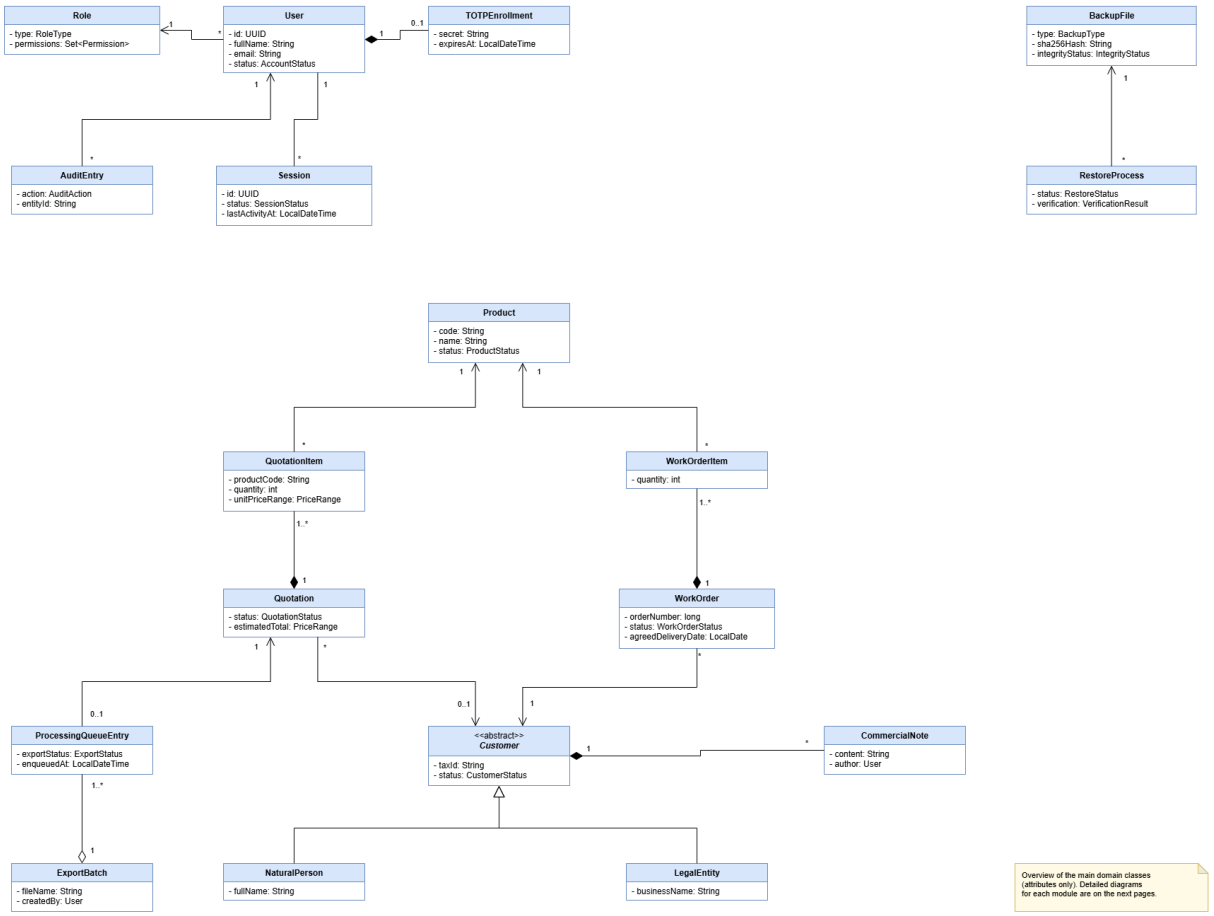
Class Diagram

The UML class diagram was developed using draw.io and is structured into twelve sections, numbered from 0 to 11. These sections are organized as follows: 0 - Domain Overview, 1 -

Authentication & Sessions, 2 - Users & RBAC, 3 - Backup Creation & Retention, 4 - Backup Inventory & Restore, 5 - Product Catalog, 6 - Customers & Commercial Notes, 7 - Quotations: Domain & Review, 8 - Chatbot & Quotation Request, 9 - Work Orders, 10 - Export Queue & Accounting, and 11 - Shared Kernel.

The primary section is 0 - Domain Overview, which provides a comprehensive global view of the domain. This section outlines the system's core entities -such as Role, User, TOTPEnrollment, Session, BackupFile, RestoreProcess, AuditEntry, Product, Customer, NaturalPerson, LegalEntity, CommercialNote, Quotation, QuotationItem, WorkOrder, WorkOrderItem, ProcessingQueueEntry, and ExportBatch- along with the structural relationships between them. The remaining eleven sections delve into the technical and operational specifics of each individual subdomain. An image representing this initial section is attached.

Domain Overview

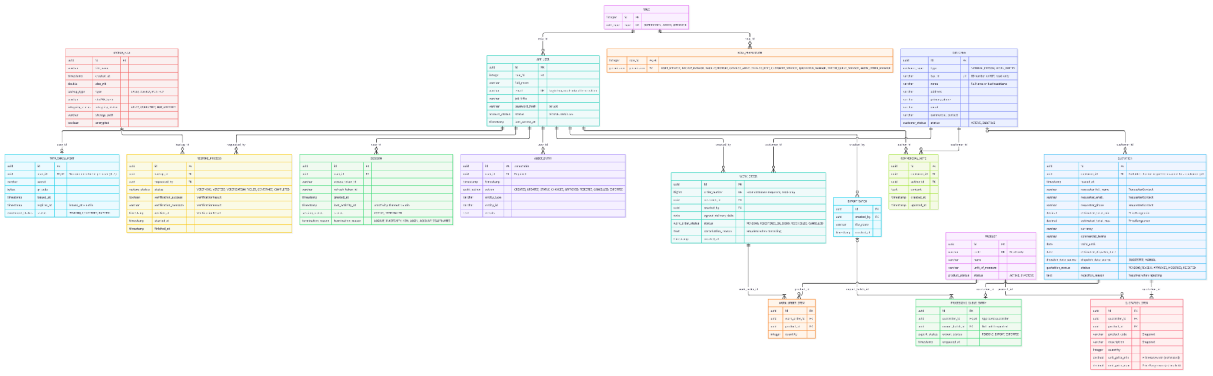


To view and analyze the complete diagram, you can access it directly via this hyperlink. Additionally, the .drawio file is available in the project repository and simply needs to be opened in draw.io.

Relational Database Model

To structure the system's data model, we designed a relational database model specifically optimized for PostgreSQL. This relational schema centralizes information across the application's various modules, defining the necessary tables and dependencies for access control (users, roles, permissions, TOTP enrollments, and sessions), corporate audit logging, and backup and restore management, as well as the core business operations: managing customers and their commercial notes, product catalogs, the workflow for quotes and work orders, and finally, the batch export queue for Siigo integration.

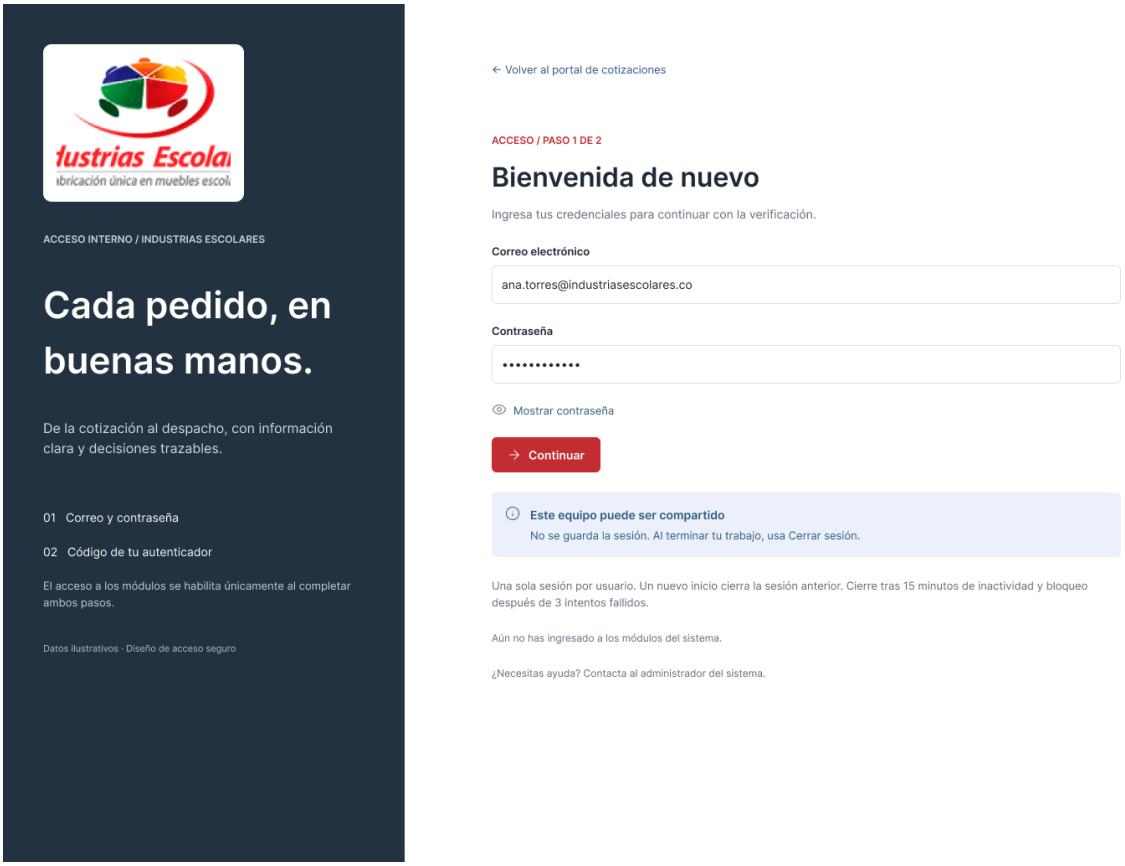
The complete diagram, along with its Mermaid source code, is attached in this same directory within the project repository. Furthermore, this diagram can be accessed via Mermaid using the provided hyperlink.



Mockups

The Industrias Escolares management system has three fixed user profiles: Super Administrator, who is Carlos Ríos; Administrator, who is Ana Torres; and Operator, who is Marta Ruiz. Since the equipment may be shared, all screens remind users to log out when they leave.

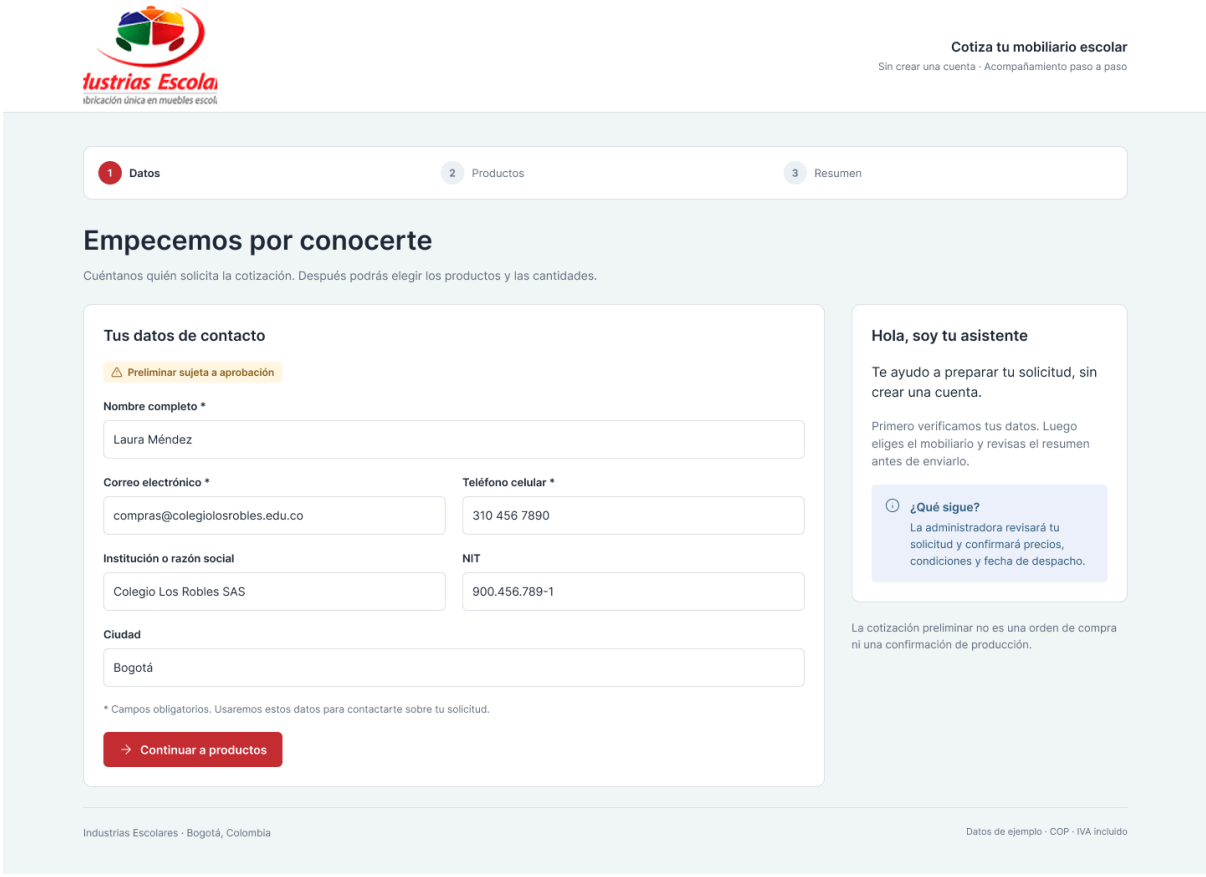
Login

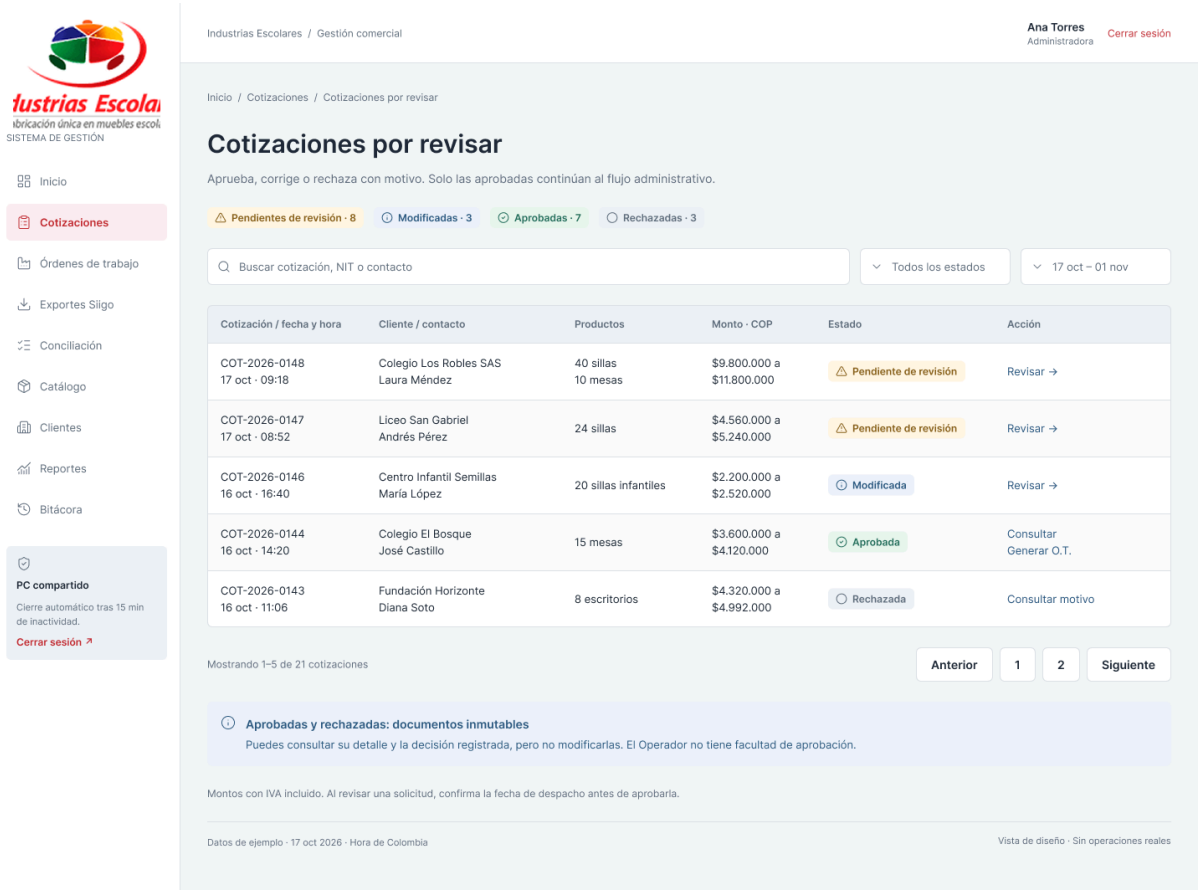


Access happens in two steps. First the person enters an email and password. Then they open their authenticator app and type the six digit code, which never arrives by email or text message. If the code is wrong or has expired, they can wait for the next one and try again.

On a first login there is an extra screen to link the authenticator, either by scanning a QR code or by typing a key. If that linking code expires, a new one is generated. After three failed attempts the account is locked and the person has to ask the administrator for help. The design also covers other cases: a session that expires after fifteen minutes of inactivity, a session closed because someone signed in from another device, and a notice that the role lacks permission for an action.

Quotations

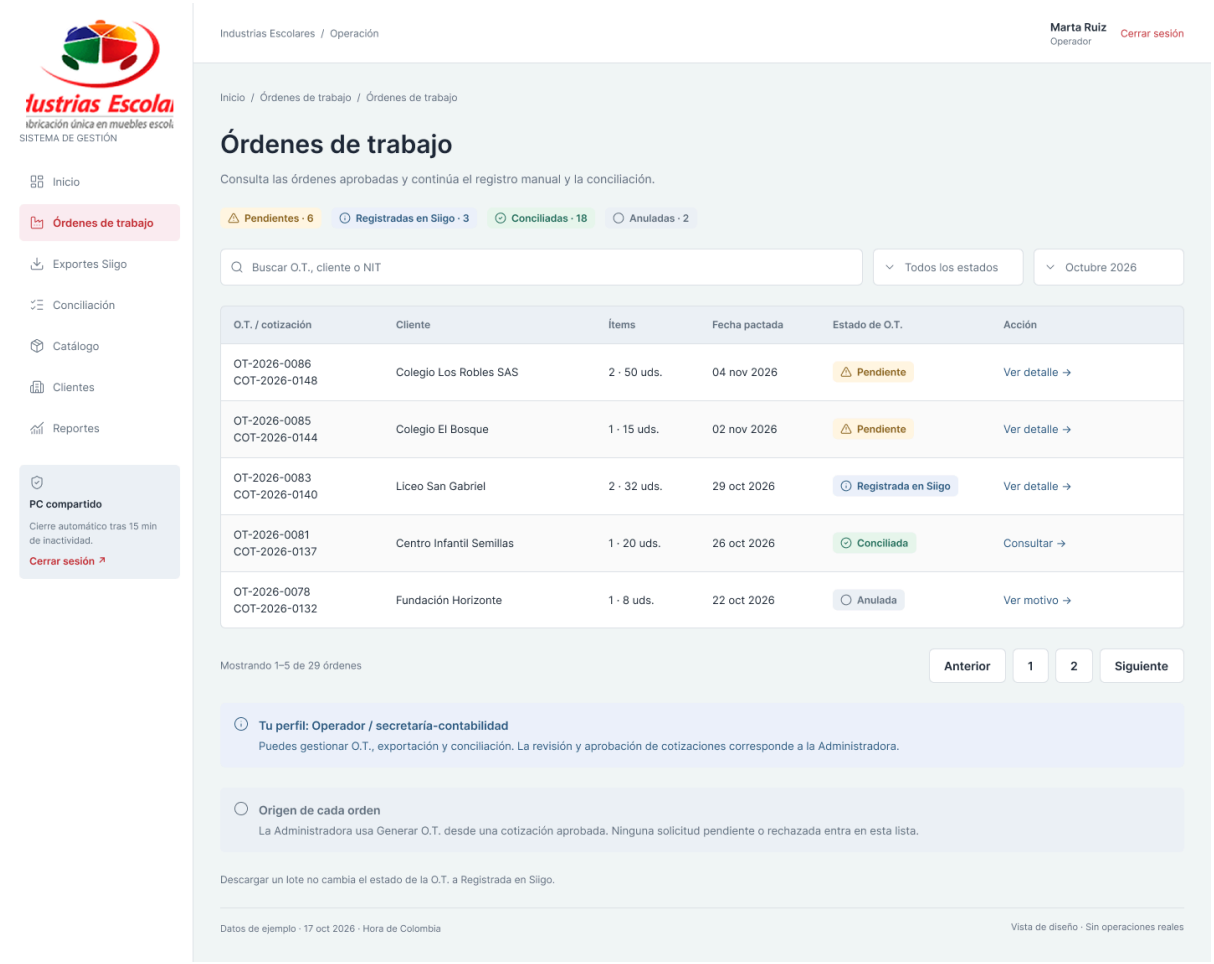

Cotiza tu mobiliario escolar
Sin crear una cuenta · Acompañamiento paso a paso
1 Datos
2 Productos
3 Resumen
Empecemos por conocerte
Cuéntanos quién solicita la cotización. Después podrás elegir los productos y las cantidades.
Tus datos de contacto
Preliminar sujeta a aprobación
Nombre completo *
Laura Méndez
Correo electrónico *
compras@colegiolosrobles.edu.co
Teléfono celular *
310 456 7890
Institución o razón social
Colegio Los Robles SAS
NIT
900.456.789-1
Ciudad
Bogotá
* Campos obligatorios. Usaremos estos datos para contactarte sobre tu solicitud.
Continuar a productos
Hola, soy tu asistente
Te ayudo a preparar tu solicitud, sin crear una cuenta.
Primero verificamos tus datos. Luego eliges el mobiliario y revisas el resumen antes de enviarlo.
¿Qué sigue?
La administradora revisará tu solicitud y confirmará precios, condiciones y fecha de despacho.
La cotización preliminar no es una orden de compra ni una confirmación de producción.
Industrias Escolares · Bogotá, Colombia
Datos de ejemplo · COP · IVA incluido



This area has two sides. The client, with no need to create an account, moves through three steps. First they leave their contact details and those of their institution. Then they choose products and quantities, while a side panel keeps showing their selection and an estimated price range. Finally they get a preliminary quotation with its reference number, which they can download as a PDF or receive by email. The summary shows the estimated range, a tentative dispatch date and the preliminary payment condition of a 30% advance. All of it is subject to approval, so prices, conditions and dispatch date are not final until the formal quotation arrives.

On the internal side, the Administrator has a tray of requests that can be filtered by status: pending, modified, approved and rejected. The amount for each request appears as a range. When she opens one, she can correct quantities, unit prices and commercial conditions. The system shows a suggested unit price for each product as a reference, but the final price is her decision. Any correction requires a written reason, and she must confirm the dispatch date before approving. She has three outcomes: approve, save a correction, or reject with a reason and an extra confirmation step. Once a quotation is approved or rejected it can no longer be changed. Only an approved one allows the work order to be generated.

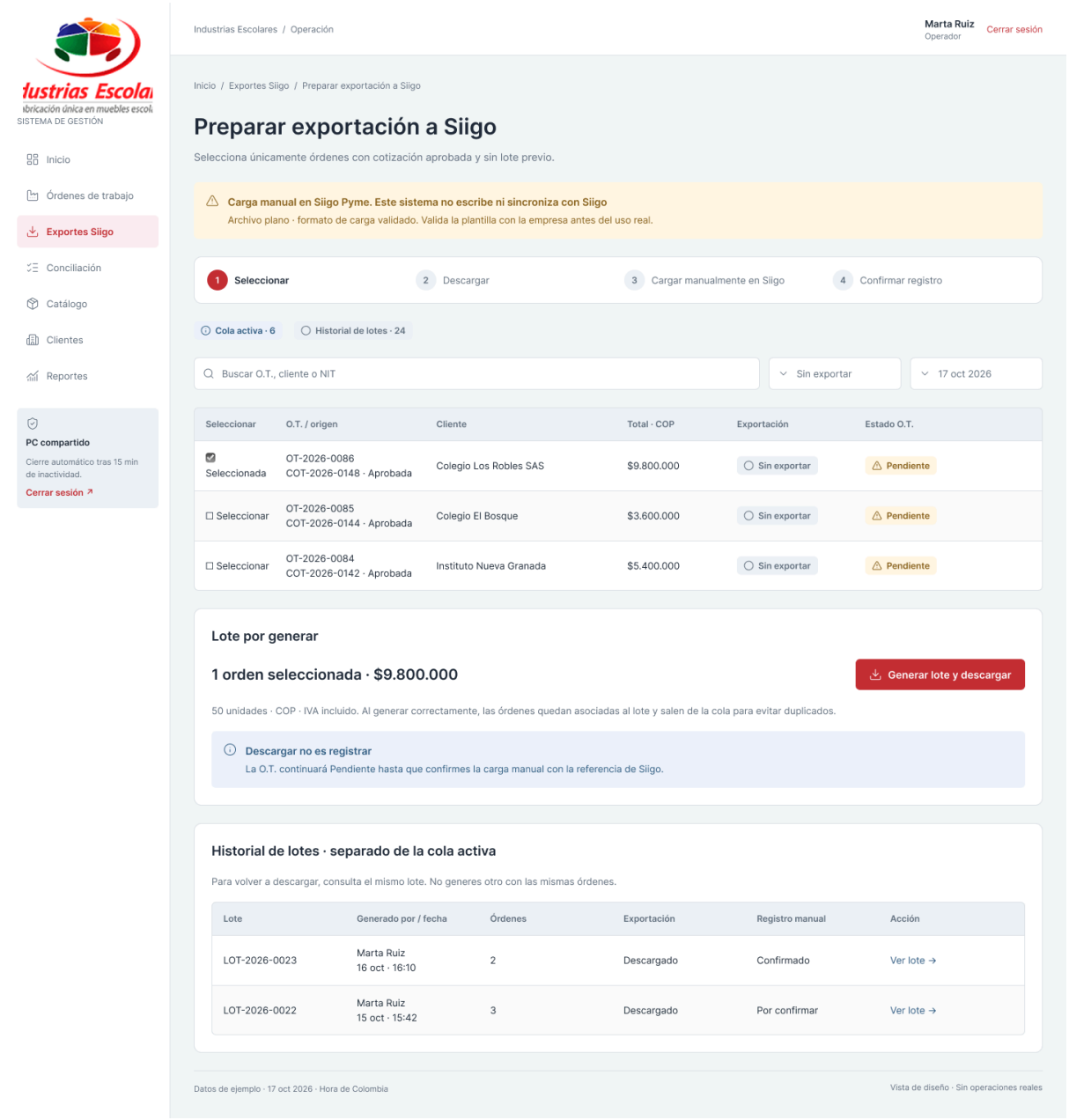
Work Orders



The Operator sees a list of all orders with their client, number of items and agreed date. Each order moves through three statuses in sequence: Pending, Registered in Siigo and Reconciled, and there is also a Voided status. Skipping statuses is not allowed. The detail view shows where the order came from, the client data, the products with their manufacturing specifications, the total, the 30% advance and the outstanding balance. All amounts come from the approved quotation.

The Administrator generates the order only once from an approved quotation, and the system prevents a second order for the same quotation. While the order is Pending or Registered, it can be edited or voided as long as a reason is recorded. Once it is Reconciled or Voided, it becomes read only.

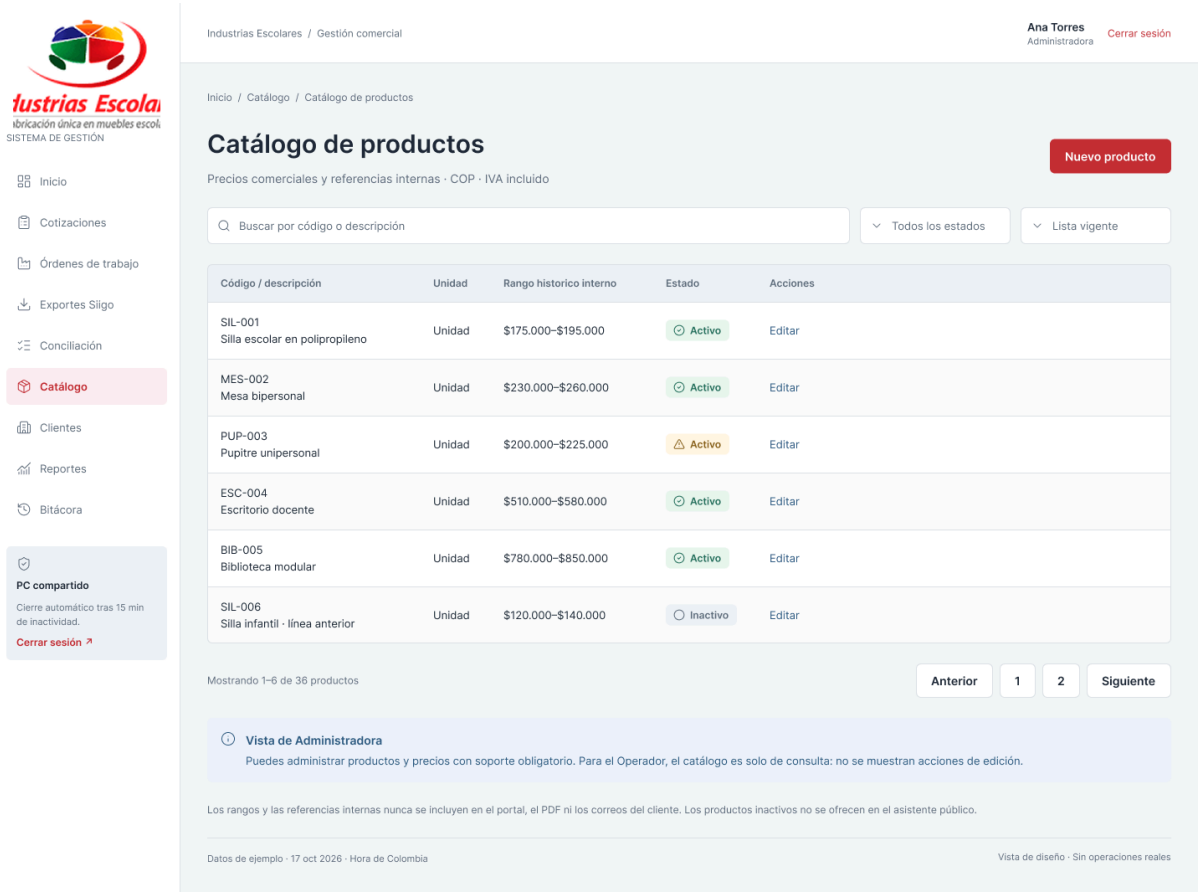
Exports to Siigo



This module does not connect to Siigo. Registration in Siigo is entirely manual, and the screen says so from the start. The Operator selects approved orders that do not yet belong to a batch, generates the batch and downloads a flat file. They then upload that file on their own in Siigo Pyme. When finished, they return to the system, enter the reference of the document they created in Siigo, check that they verified the amounts and confirm the registration.

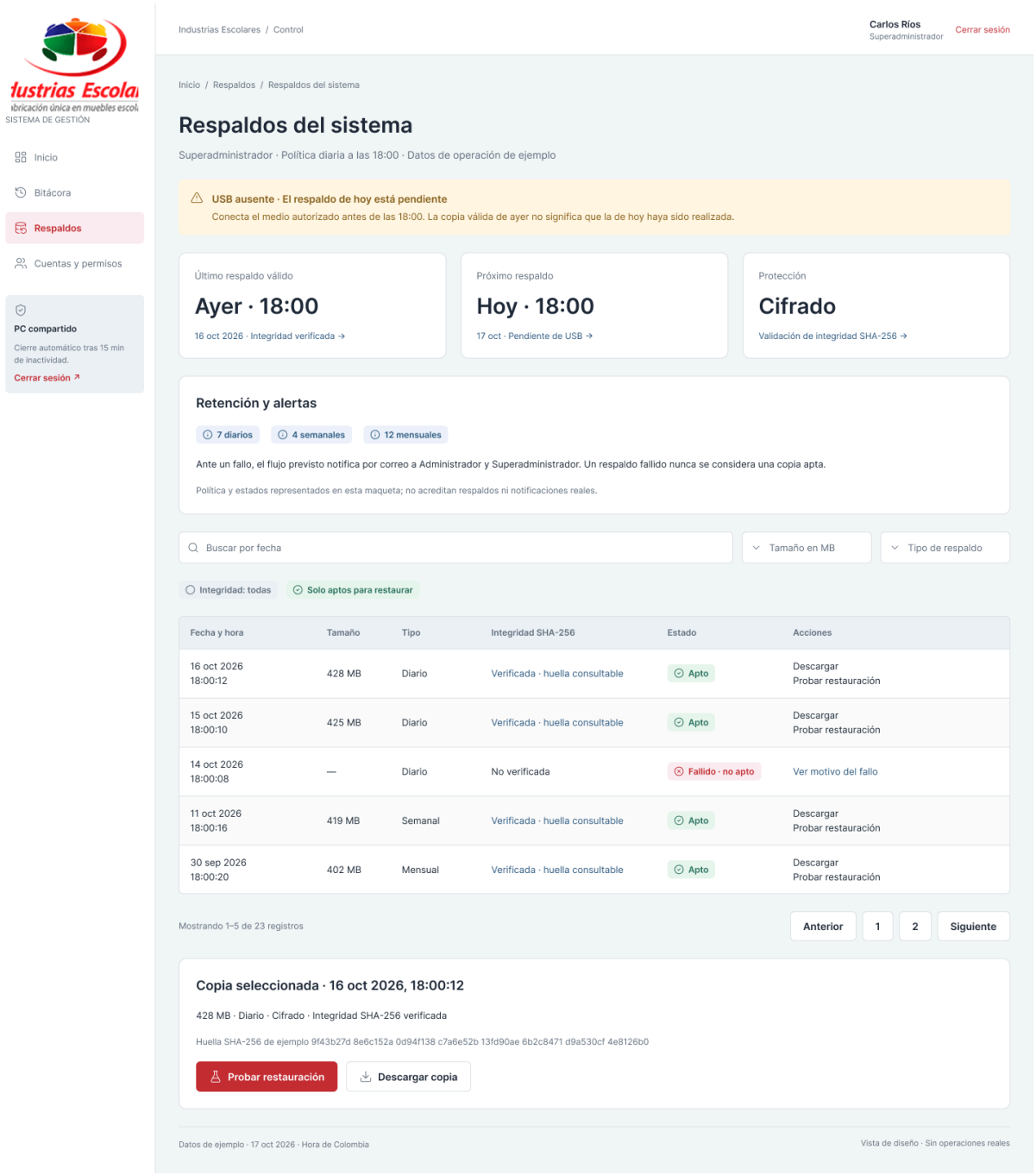
Downloading the file does not change the order's status. Only that later confirmation marks it as Registered. A batch history is kept apart from the active queue, so a batch can be downloaded again without generating a new one with the same orders. If file generation fails, no order is marked or removed from the queue, and the Operator can simply retry.

Catalog



The list shows each product with its code, description, unit, internal historical price range and status, with search and filters. The Administrator can create new products and edit existing ones, but editing only changes product data. Prices are not set or changed from here, and there is no price change history. The historical range is there only as a reference, and it never appears in the client portal, in PDFs or in emails. Inactive products are not offered to the public. The Operator can only view the catalog.

Backups



This section belongs to the Superadministrator. The panel shows the last valid backup, when the next one will run, that the copy is made at six in the evening to a USB drive, and that it is encrypted. If the USB is not connected, an alert appears, because yesterday's valid copy does not guarantee that today's was made. Retention is seven daily copies, four weekly and twelve monthly. If a backup fails, an email notification is sent and the copy is never considered valid. The table lets the user search and filter copies, check their integrity, download them or test a restore.

Restoring is a four step process. The user chooses a copy, runs an isolated test in a separate environment that does not touch production, reviews the result and finally confirms

in production. That last confirmation requires checking an authorization box and typing the word RESTAURAR. If the test fails, the restore is cancelled and production stays untouched.

Other Views

The Administrator's home screen gathers what is pending for the day: quotations to review, orders to register and orders to reconcile. It also includes a five step summary of the whole process and a weekly report of requests.

Reconciliation compares an order against the Siigo document line by line, covering quantities, subtotals and the total. To reconcile, the person must attach the evidence, write an observation and check their verification box. The order then becomes read only and cannot go back. If differences are found, the final action stays disabled until they are resolved.

The client 360 view lets the user search by tax ID or company name and review each client's history of quotations and orders. The tax ID and the type of person cannot be changed. Clients are never deleted, only deactivated, and they keep their history. There are also private notes that never appear in PDFs or emails, and an alert when one client accounts for a very large share of sales.

The activity log is read only for every profile. It records who did what and when, with filters by actor, action, entity and date, and each event can open its evidence, such as an approval, a batch generation, a manual registration or a reconciliation. Finally, in accounts and permissions the Superadministrator creates, edits and deactivates accounts across the three fixed profiles, sets an initial password and lets each person link their authenticator on first login.

Overall Workflow

Everything starts when a client requests a quotation from the portal and sees an estimated range. The Administrator reviews it, adjusts the unit prices if needed, confirms the dispatch date and approves it, which enables the work order. The Operator includes the order in a batch, uploads it manually to Siigo and confirms the registration with the document reference. Finally the order is reconciled against Siigo and closed as read only. Each step leaves a trace in the activity log, and the Superadministrator takes care of backups and accounts.

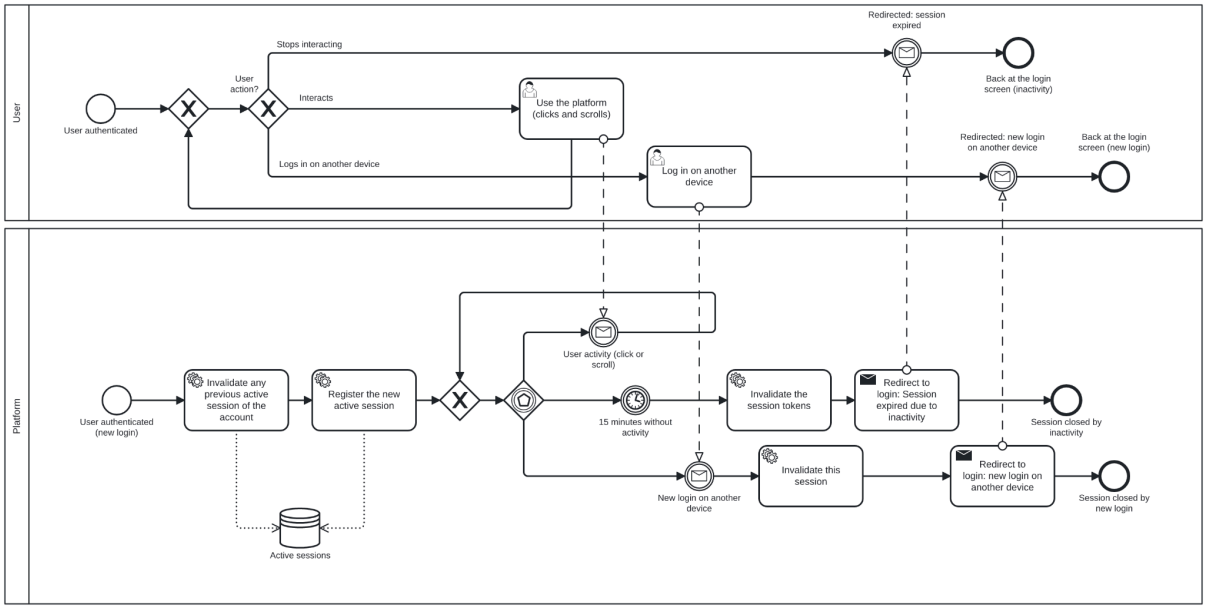
Note: *To view all the designs for the additional windows, as well as those for the other views, you can visit the following link or this other one.*

Business Model Processes

A total of 9 processes were modeled using the BPMN 2.0 standard, divided into 3 large and 6 small ones. The characteristics and workflow of each are detailed below.

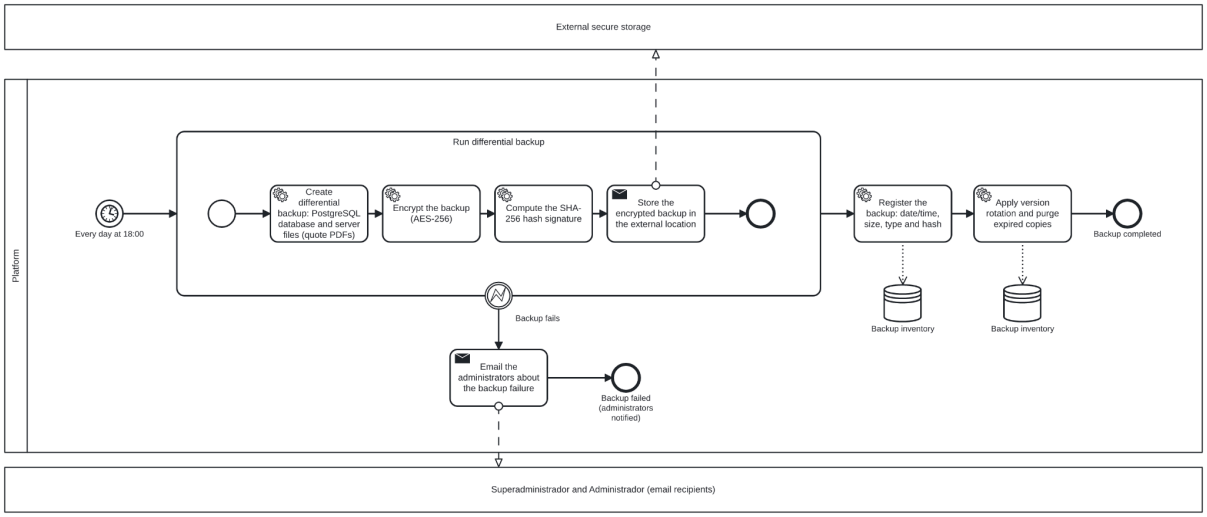
Session Management

The first process corresponds to session management, designed to monitor inactivity and prevent simultaneous access. Upon logging in, the system ensures a single active access by invalidating any previous connection from the same account. The session is kept alive by user interaction, but it automatically closes -redirecting to the login screen- if 15 minutes of continuous inactivity are detected or if a new login from another device is registered.



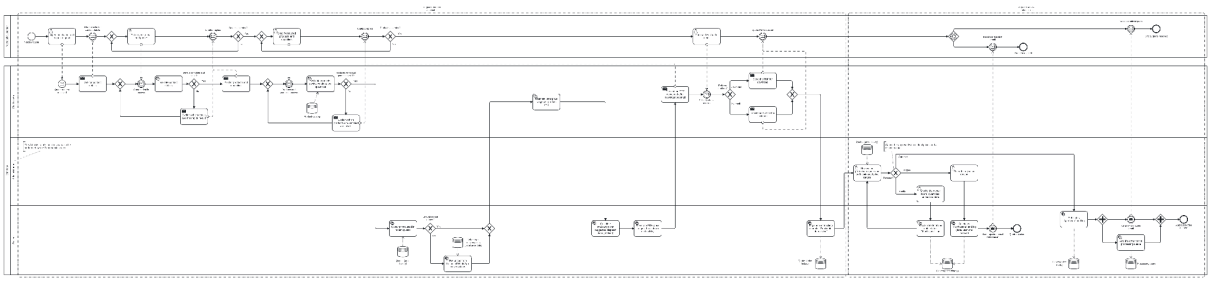
Automated Backup Management

Meanwhile, the second process describes the execution of the platform's automated backups, scheduled daily at 18:00. The system generates a differential backup of the database and files, which is encrypted with AES-256, verified via a SHA-256 hash signature, and sent to secure external storage. Finally, the backup is logged in the inventory and a version rotation is applied to purge expired copies; if the process fails at any point, an email alert is sent to the administrators.



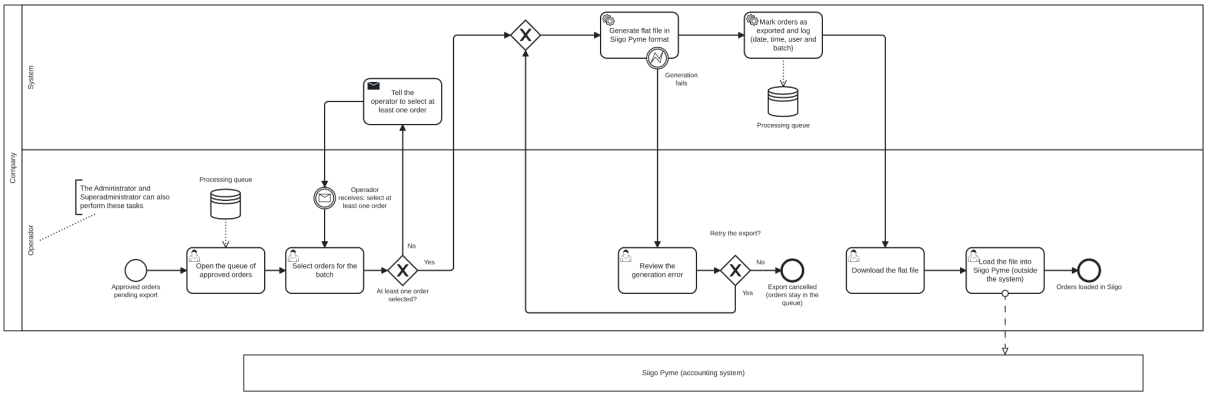
Quote Request and Approval

The third process covers end-to-end quote management, split into the customer request via an AI chatbot and subsequent administrative review. The chatbot validates the request against the catalog and estimates price ranges based on past quote history or previous WhatsApp logs. Afterward, the administrator reviews the preliminary quote to modify it, issue a justified rejection, or approve it with final pricing and a dispatch date, which queues the order and sends the official PDF quote to the customer.



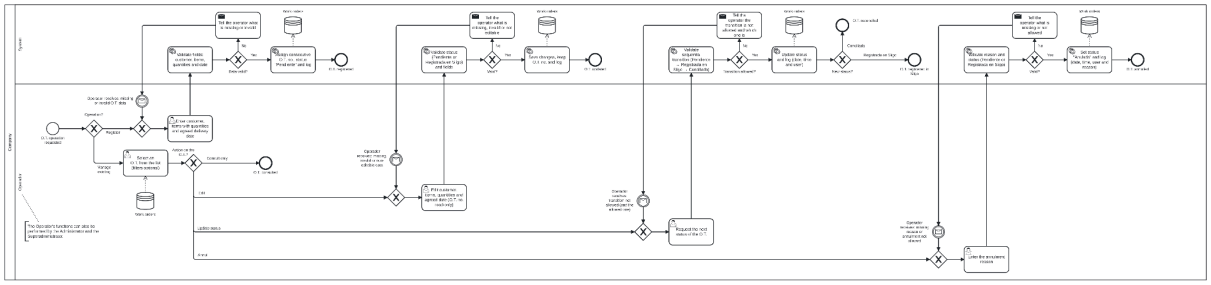
Exporting Orders to Siigo Pyme

The fourth process manages the generation of the flat file for external integration of approved orders. The operator selects a batch from the queue, ensuring at least one order is selected before generating the Siigo Pyme formatted file. Upon successful generation, the system marks the orders as exported to prevent duplicates, allowing the operator to download the file and upload it manually to the accounting system; if generation fails, the user is prompted to review the error and retry the export.



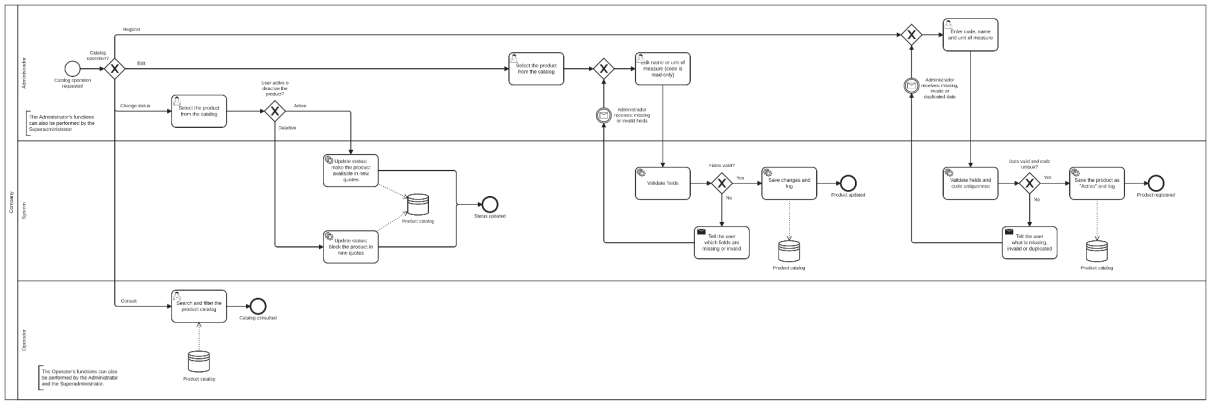
Work Order Management

The fifth process manages the full lifecycle of work orders (W.O.). Operators can register new W.O.s (validating customer, items, and date), consult them, or edit their details while preserving the assigned ID. Additionally, the system enforces sequential status transitions, progressing step by step from *Pending* to *Registered in Siigo* and ultimately to *Reconciled*, while allowing justified annulments for pending or registered orders and keeping an audit log of every change.



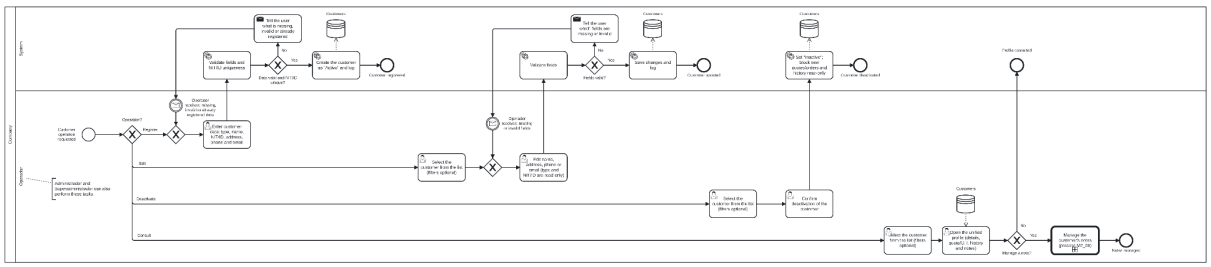
Product Catalog

The sixth process manages the product catalog under a role-based access model. Administrators can register new items ensuring code uniqueness (without an official price field), edit names or units of measure (keeping the code read-only), and activate or deactivate products to enable or block their use in new quotes. Meanwhile, operators have restricted read-only access to search, filter, and view the catalog.



Customer Management

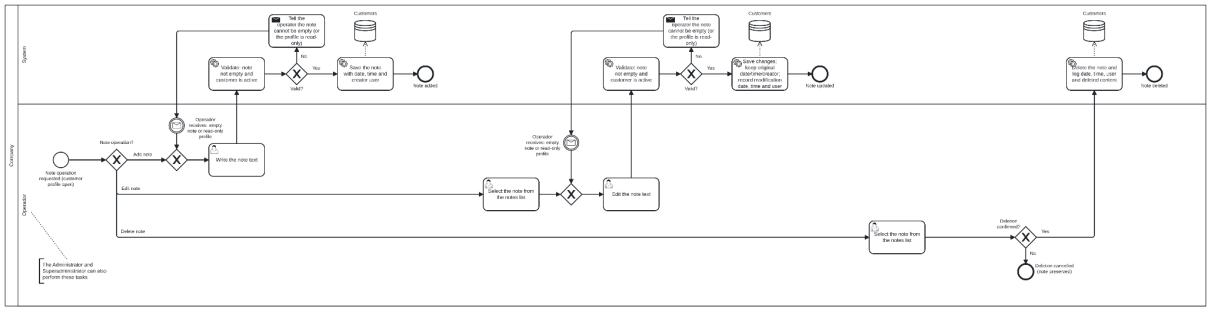
The seventh process covers the administration of customer profiles. It allows registering new customers while enforcing unique NIT/ID document numbers, as well as editing contact details while keeping the ID read-only. It also handles logical deactivation to prevent new quotes or orders while keeping past records read-only, and unified profile consultation, which allows invoking the customer notes sub-process.



Customer Notes Management

The eighth process operates as a customer profile sub-process to manage commercial notes and logs. It allows adding new notes while ensuring the customer profile is active and

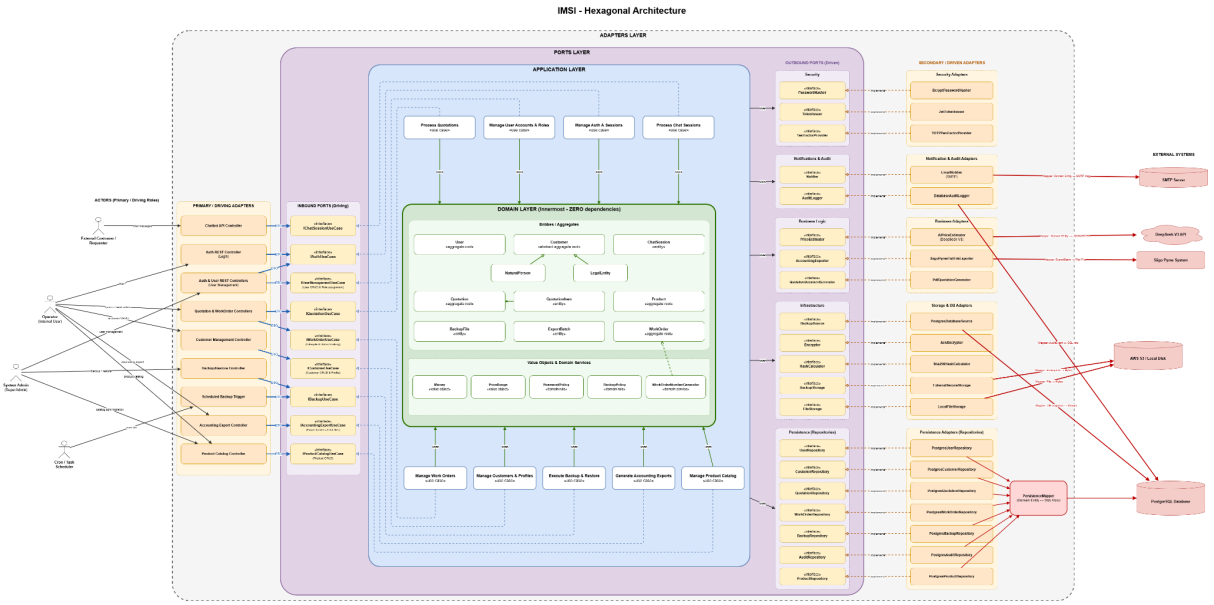
the text is not empty, editing existing notes preserving original creation details while auditing changes, and deleting notes upon user confirmation, logging the deleted text, date, time, and user in the system audit.



For a clearer and more detailed view of these diagrams, you can access them via this link; additionally, all process files are available directly within the dedicated folder in the project repository.

Architecture Diagram

For the system's technical design, we chose to implement a Hexagonal Architecture. This framework, which was carefully designed using the draw.io tool, allows the core business logic to be completely isolated from any external technology, ensuring that the domain layer operates with zero dependencies. In this way, the system is organized through ports and adapters that securely and structurally connect the application's core with databases, user interfaces, and third-party services.



Additionally, the original and interactive diagram can be explored in depth at this link, and the file can also be found inside the repository folder.

Declaration of the Use of Artificial Intelligence

For the preparation of this document, we declare that the use of Artificial Intelligence tools -specifically Gemini and Claude- was strictly limited to support, review, and translation purposes. First, these tools were used exclusively to translate the system domain, business rules, and technical descriptions into English. Likewise, we relied on them to conduct brainstorming sessions to explore new proposals and refine our design components, a process that was carried out only after we had independently defined and structured the system's core functionalities, database models, mockups, business model processes, and the software architecture diagram.

Furthermore, Gemini and Claude were employed as a validation mechanism to ensure there were no weak points or gaps in the project's overall technical reasoning. This included reviewing the consistency between the relational database model, the UML class diagrams, the business rules, the proposed mockup workflows, the core business process flows, and the technical interactions, layers, and communication flows detailed in the architecture diagram to guarantee their logical soundness and architectural integrity. Finally, we used both artificial intelligences to confirm that each of these design elements strictly complied with the correct structure and formal presentation required for this academic submission.